

Active Proprioceptive Recognition of Object Shape and Size with a LEAP Hand: A Simulation-Based Study

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Abstract

Can the final joint configuration of one controlled grasp identify both the shape and size of a known object type at an unseen scale? We investigate this question with a 16-joint LEAP Hand and a reproducible simulation-to-hardware pipeline. Three fixed computer-aided-design objects—a cube, sphere, and cylinder—are placed at one common pose in MuJoCo and uniformly scaled from 50% to 150% in 5% increments. During each grasp, collision stops all four joints of a contacted finger. The resulting 16 final joint angles train a small neural network to classify shape and regress scale; a separate rule rejects ambiguous or distant grasps as unknown. On 15 held-out simulated sizes, the network correctly classified 13 (86.7%) and achieved a 1.02 mm mean absolute ground-truth-dimension-weighted scale error. The rejection rule accepted three of these 15 grasps, all correctly classified. The physical-hand program implements the corresponding grasp using motor current as a contact proxy, but this manuscript reports no physical-hand recognition trials. The results establish a simulation baseline for joint-angle-only recognition under fixed placement, while leaving real-world transfer and rejection of genuinely novel objects to future experiments.

Keywords: proprioception, active perception, robotic hand, object recognition, size estimation, simulation-to-real transfer

1. Introduction

A robot can obtain information about an object by acting on it. For a dexterous hand, the joint configuration reached while closing around a surface depends on both the object’s geometry and its size. This suggests a form of active perception in which the movement of the hand generates the observation used for recognition. Motor positions are available on the

LEAP Hand without an external camera or dedicated tactile skin (Shaw et al., 2023).

Prior work demonstrates that hand kinematics can support recognition. Vásquez and Perdereau (2017) constructed proprioceptive shape signatures from joint angles and studied recognition across object sizes and poses. Sintov and Meir (2023) used computationally generated grasps for kinesthetic object recognition and evaluated recognition of both individual instances and broad geometric categories. These studies motivate a controlled experiment that asks whether a compact final-angle observation can support *both* shape classification and a continuous estimate of physical scale.

The research question here is deliberately narrow: given a fixed object mount, a fixed exploratory grasp, and a set of three trained object classes, how well can the final 16 joint angles distinguish class and estimate size at scales excluded from training? The fixed mount removes object-pose variation so that the effect of scale can be examined first. The project also implements a matching grasp and prediction program for the physical hand. Its intended transfer mechanism replaces simulated mesh contact with a motor-current threshold, while preserving the four-joint finger stopping rule and final-angle input. This transfer mechanism has not yet been evaluated with labeled real-hand trials.

The contributions of the present implementation are (i) a parameterized MuJoCo grasp-data generator for STEP objects and a 16-joint LEAP Hand model, (ii) a reproducible joint-angle classifier and size regressor with an explicit unknown-output rule, and (iii) a physical-hand execution and logging program. The measurements reported below concern the simulated data generator and held-out scale evaluation.

2. Materials and Methods

2.1. *Hand, software, and object models*

The platform is a four-finger, 16-actuator LEAP Hand (Shaw et al., 2023). The logical joint vector is ordered as index joints 0–3, middle joints 4–7, ring joints 8–11, and thumb joints 12–15. All commanded and recorded joint angles in the experiment are in degrees. MuJoCo stores joint coordinates internally in radians (Todorov et al., 2012). The included hand URDF and link meshes define the simulated geometry. The URDF limits at joints 0, 8, 13, and 14 were expanded to admit the configured grasp; no simulation-to-hardware angle offset is currently applied.

Three STEP models are used: a cube, sphere, and cylinder. The reference cube side length is 50 mm. The reference diameter is 50 mm for both the sphere and cylinder. Every object is fixed at the same pose relative to the simulation frame: translation $(-0.060, -0.037, -0.035)$ m and roll-pitch-yaw orientation $(-180^\circ, 0^\circ, 0^\circ)$. STEP coordinates are treated as millimetres, converted to metres, and multiplied by a uniform scale factor. The same factor changes every object dimension, although the model predicts only the designated size measure in Table 1.

Table 1: Simulated object classes and the physical quantity predicted. Each class has 21 scales from 50% to 150%, inclusive.

Class	Predicted quantity	Reference	Sweep range
Cube	Side length	50 mm	25–75 mm
Sphere	Diameter	50 mm	25–75 mm
Cylinder	Diameter	50 mm	25–75 mm

2.2. A standardized active grasp

The grasp uses the same start posture, target posture, and one-degree command increments for every simulated object. Table 2 specifies the joint vectors in the logical order defined above. Joint values are clamped at their target angles. The grasp terminates when all four fingers have stopped, with a cap of 250 iterations. A finger stops as one unit: contact at any of its links freezes all four joints of that finger, while other fingers may continue closing. Reaching all configured target angles also stops a finger.

Table 2: Configured grasp angles and per-step increments, in degrees. Each bracket contains the four joints of one finger.

Finger	Start	Target	Increment
Index	[0, 0, 0, 0]	[−75, 90, 90, 60]	[−1, 1, 1, 1]
Middle	[0, 0, 0, 0]	[0, 80, 80, 65]	[1, 1, 1, 1]
Ring	[0, 0, 0, 0]	[75, 90, 90, 60]	[1, 1, 1, 1]
Thumb	[0, 180, 40, 25]	[0, 100, 110, 80]	[1, −1, 1, 1]

2.3. Simulation of contact and data collection

The simulation code imports each STEP object with CadQuery, tessellates it into a triangle mesh, and mounts that mesh as a fixed body in a MuJoCo scene containing the hand URDF. Explicit object–hand collision pairs are added. Contact geometries are assigned to a finger through their parent bodies; contact with the palm or an unexpected stationary finger is treated as an invalid scene. Any object already touching the hand in the initial posture is also rejected.

At each grasp iteration, the simulator advances one active finger at a time and tests the proposed posture for mesh contact. If contact occurs, eight bisection steps between the last clear posture and the blocked posture locate a nearby collision-free final angle. The entire finger then stops. This is a kinematic contact experiment: joint positions are set directly, with no learned grasp policy, motor dynamics, force prediction, or simulated motor current. The recorded feature for scene i is the final vector

$$\mathbf{q}_i = (q_{i,0}, \dots, q_{i,15}) \in \mathbb{R}^{16}, \quad (1)$$

along with its class, scale, physical size, and each finger’s stop reason. A complete step-by-step trace is saved for replay in the MuJoCo viewer.

The scale sweep is $s \in \{0.50, 0.55, \dots, 1.50\}$, so the three classes produce $3 \times 21 = 63$ simulated scenes. Each class has exactly one deterministic simulated grasp per size. No object-pose randomization or repeated physical trials are part of this dataset.

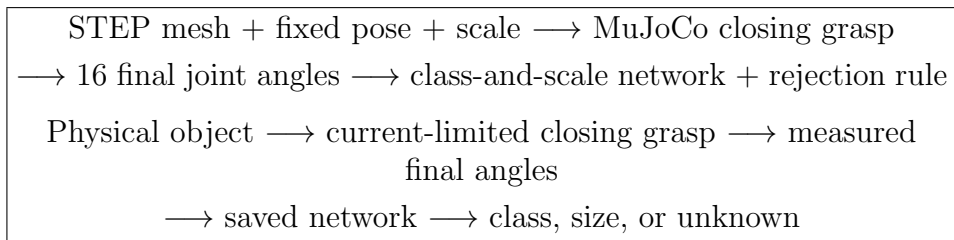


Figure 1: Implemented data and inference paths. The lower path is implemented for the real hand but has not supplied the quantitative results in this paper.

2.4. Feature preparation and model

Only the final angles in Eq. (1) are model inputs. Positions, velocities, motor currents, fingertip trajectories, and per-step contact histories are not

training features. Each angle is normalized by the configured grasp span:

$$x_j = \begin{cases} \frac{q_j - a_j}{b_j - a_j}, & b_j \neq a_j, \\ 0, & b_j = a_j, \end{cases} \quad (2)$$

where a_j and b_j are the start and target angles. A fixed joint has no simulated variation from which to identify an object. The input $\mathbf{x} \in \mathbb{R}^{16}$ passes through a fully connected 32-unit hyperbolic-tangent layer. The four output values consist of three class logits and one continuous predicted scale factor $\hat{s}$. A softmax converts the logits to class probabilities. With three classes, the network has $16(32) + 32 + 32(4) + 4 = 676$ trainable parameters.

Training minimizes mean class cross-entropy plus mean squared scale error:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log p_{i,c_i} + \lambda \frac{1}{N} \sum_{i=1}^N (\hat{s}_i - s_i)^2, \quad \lambda = 1. \quad (3)$$

The NumPy implementation uses full-batch Adam for 1000 epochs, a learning rate of 0.02, and random seed 7. For each simulated grasp, 50 independent copies receive zero-mean Gaussian measurement noise with a 0.5-degree standard deviation per angle; the unmodified example is retained. The augmented copies approximate position measurement variation but do not model systematic calibration error.

2.5. Validation protocol and unknown decision

The data split is made by *size before augmentation*. Every fifth scale—50%, 75%, 100%, 125%, and 150%—is held out for each class. Thus 48 distinct simulated scenes produce 2448 training examples (48 original vectors plus 50 noisy copies each), while 15 unmodified scenes are used for validation. A separate final model is then trained on all 63 scenes and their noisy copies (3213 examples) for later physical-hand use. The validation figures in this paper refer to the model trained on the 48-scene subset.

Raw classification accuracy is the fraction of the 15 validation scenes whose highest-probability class equals the ground-truth class. For scale, the reported mean absolute error is

$$\text{MAE}_{\text{size}} = \frac{1}{15} \sum_{i=1}^{15} |\hat{s}_i - s_i| d_{c_i}, \quad (4)$$

where d_{c_i} is the *ground-truth class's* reference dimension. This measures scale regression in physical units even for a misclassified scene; it is not the error of the final class-conditioned size output when the predicted class is wrong.

The deployed output may instead be **unknown**. Four checks are applied in order. First, the Euclidean distance from the normalized input to its nearest simulated reference vector must not exceed a distance threshold. That threshold is 1.5 times the largest nearest training-scene distance of the 15 held-out scales (0.0754 for the present dataset). Second, the largest class probability must be at least 0.75. Third, the network class must agree with the class of the nearest simulated reference vector. Fourth, the difference between the two smallest *classwise* nearest-reference distances, measured in raw angle degrees, must be at least 2° . A failure at any step returns **unknown** without a size. Otherwise, the predicted scale is clipped to $[0.50, 1.50]$ and multiplied by the reference dimension for the predicted class.

The distance threshold was calculated from the same 15 scales on which accepted/rejected outcomes are reported. Those outcomes are therefore descriptive, not an independent evaluation of the unknown-object rule. No untrained CAD object class was tested.

2.6. Physical-hand implementation and reproducibility

The hardware program sends the grasp through **ExArm**. It reads all 16 motor positions, velocities, and signed currents through the Dynamixel driver. For each moving joint, a current exceeding its calibrated threshold in the commanded direction stops that joint's entire finger at its measured pose. The per-joint threshold vectors are $[22, 80, 25, 20]$ for each of the index, middle, and ring fingers and $[25, 70, 25, 20]$ for the thumb, in the values returned by the motor-state reader. The hand settles for 2s before the grasp, waits 0.2s between command steps, and aborts after repeated state-read failures or more than 250 steps. These thresholds are configurable and may require recalibration; motor current is a contact proxy, not a direct measurement of contact force.

After the final state read, the physical program supplies the measured 16-angle vector to the saved model, prints class and size or an unknown reason, and saves a timestamped JSON result. It also saves joint-state traces and the final angle vector as CSV files. This code path is implemented, but the repository contains no labeled real-object results, so no sim-to-real recognition accuracy is claimed.

The experiment settings are in `utils/Constants.py`. The main entry points generate and train simulated grasps (`Train_Proprioception.py`), replay saved grasp trajectories (`View_Proprioception.py`), and run recognition on the physical hand (`Run_Proprioception.py`). The `Data/Proprioception` directory holds final grasp angles, step traces, augmented samples, validation rows, the training report, and the saved model weights. The implementation and data form part of the Ex-ARM project (Pranav, 2026).

3. Results

The generated dataset contains 63 completed simulated grasps and 5004 recorded time-step rows, including the initial posture of each scene. Across the 252 finger endings (four per scene), 241 were attributed to object contact and 11 to reaching the configured angle limits. These counts confirm that the simulated closing motion generated object contact in many finger trajectories; they are not a measure of recognition performance.

On the 15 scales excluded from the validation model’s training set, 13 raw network class predictions were correct (86.7%). Table 3 shows results by class. Both raw errors were sphere-or-cylinder scenes predicted as cube. The scale-factor regression achieved the ground-truth-dimension-weighted MAE of 1.02 mm defined in Eq. (4).

Table 3: Held-out simulated-scale validation. Accepted means that none of the four unknown-output checks rejected the grasp. The acceptance rule uses a distance threshold derived from these held-out scales.

True class	Scenes	Raw class correct	Accepted (correct)
Cube	5	5	1 (1)
Sphere	5	4	1 (1)
Cylinder	5	4	1 (1)
Total	15	13	3 (3)

The unknown-output rule rejected 12 of the 15 held-out grasps: four because the two nearest classes had similar angle vectors, seven because the network’s maximum class probability was below 0.75, and one because the network and nearest reference class disagreed. All three accepted grasps had the correct class, but the acceptance rate was only one fifth. Since the

threshold was set using these same held-out sizes and no novel object class was tested, this result cannot establish real-world unknown-object detection.

4. Discussion

The held-out-scale results support the limited proposition that a single controlled closing motion can carry information about both class and scale when the object remains at a known pose in the simulator. At the same time, some sphere and cylinder configurations produced final angles that the network assigned to the cube class, and the conservative rejection checks discarded most validation grasps. A larger set of independent test sizes, repeated grasps, and baseline classifiers would be required to estimate robustness and compare alternative representations.

The central transfer uncertainty lies in how a simulated collision corresponds to a motor-current event on the physical hand. Real motors, transmission compliance, current noise, friction, and object mount errors can shift the final angle relative to the idealized collision-free boundary used for training. Gaussian perturbation of 0.5 degrees does not reproduce these effects. The experiment also holds orientation and position fixed, so its numerical results do not generalize to freely held or moving objects. Because each class has one mesh and only uniform scaling, the model has not learned within-class shape variation.

The present manuscript concerns the implemented final-angle method. It does not use fingertip reconstruction, multi-grasp surface mapping, or an adaptive exploration policy. Such methods could add information where a single grasp is ambiguous, but their benefit would need to be tested against this baseline. Similarly, open-set performance requires independent objects that were absent from both training and threshold selection.

A physical evaluation should next mount measured examples of all three classes at the simulated pose and collect repeated labeled grasps over several sizes. The comparison should report class accuracy, size error, rejection rate, and differences between each real final-angle vector and its matching simulated vector. Separate tests with new shapes and perturbed placements would then test unknown detection and domain transfer without tuning thresholds on the test cases.

5. Conclusion

This study implements active object recognition with a LEAP Hand using only final joint angles. With a fixed object pose and 63 simulated grasps across cube, sphere, and cylinder models, a 676-parameter network classified 13 of 15 held-out sizes correctly and achieved 1.02 mm mean absolute scale-regression error in the reported size units. A rule-based unknown output accepted three held-out grasps, all correctly classified, at low coverage. These are simulation results. The corresponding physical current-limited grasp and inference code are available for experimental measurement, but the accuracy of physical recognition and the ability to reject genuinely novel objects remain open questions.

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